

operability, factory test support, operator certification, device health diagnostics security, vulnerability and log analysis support.

Growth drivers

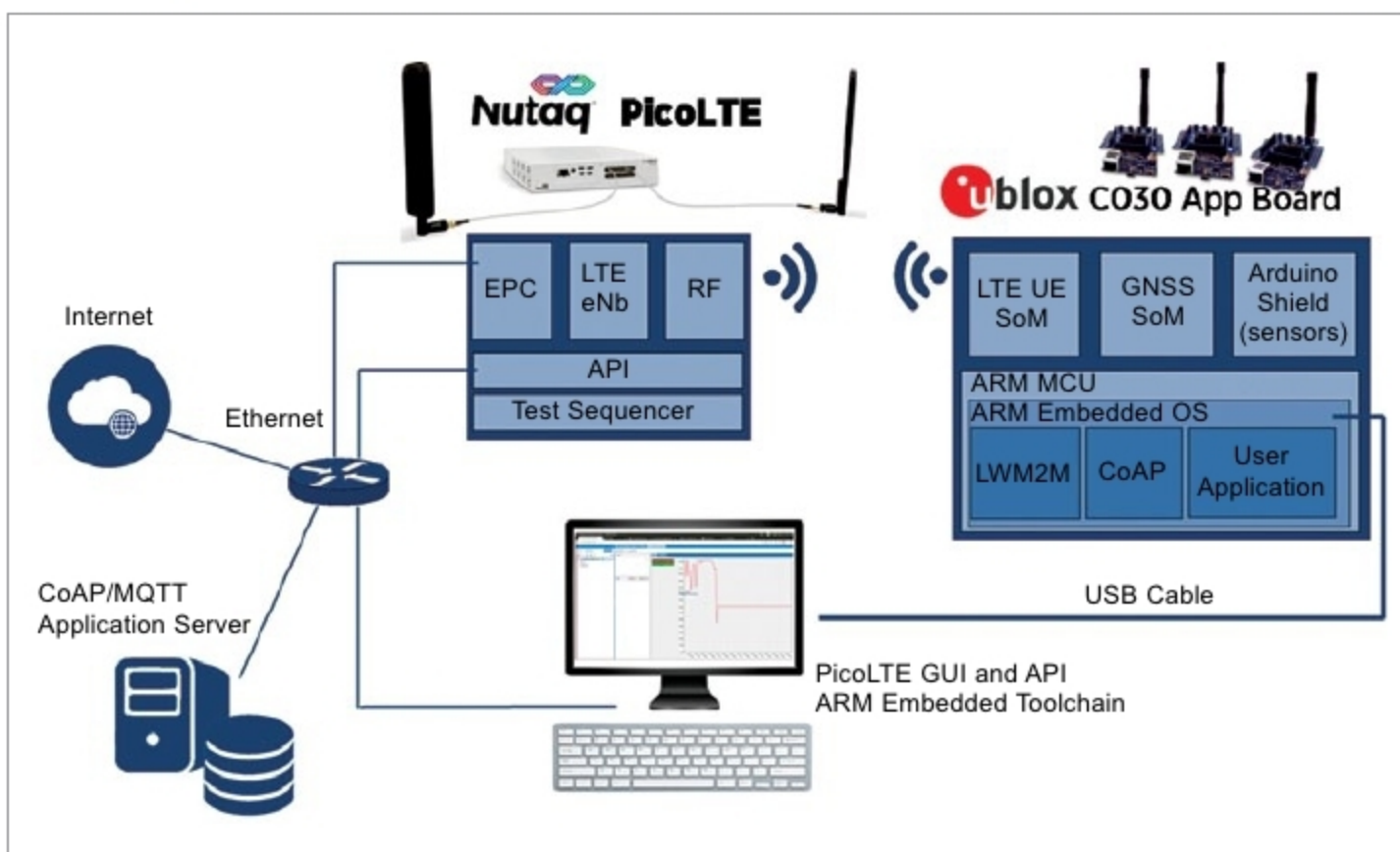
Growth drivers for test and measurement (T&M) of consumer electronics are explained below.

Automotives. Growing demand for connected cars, evolved infotainment systems and autonomous driving needs T&M equipment for the automotive industry. Adoption of wireless communications standards like Long-Term Evolution (LTE) and Wi-Fi are highlighting the need for more radio frequency (RF) testing.

Technologies related to driverless vehicles, collision avoidance and the new Wi-Fi standard 802.11p are looking to capture and analyse as much data as possible to maximise their product designs. Data analytics is a key driver in the growth of data acquisition and analysis.

Environmental testing for electronic automotive components. ISO 16750 standard covers test procedures for electrical systems and electronics for automotive applications. Testing ensures that all of the vehicle's electronic components comply with relevant regulations and electromagnetic compatibility (EMC) test standards.

Data centres. Higher transmission speeds and new standards for applications in data centres are creating the demand for high-speed digital test equipment. Demand for greater bandwidth continues to increase the need for higher transmission rates in data centres. Modulation schemes such as pulse amplitude modulation (PAM), differential quadrature phase-shift keying (DQPSK) and quadrature amplitude modulation (QAM) require more channels and upgraded instruments. Continuous evolution towards greater bandwidths and sample rates has led to the emergence of new standards.



PicoLTE IoT narrowband and Cat-M1 test and measurement platform (Credit: www.nutaq.com)

Power management applications. Power supplies are driving the demand for electronic test equipment as engineers strive to maximise their systems' efficiency. Rapid advances in new applications like electric vehicles (EVs), energy storage, wireless devices and wireless power transfer are also driving the demand for high-resolution instruments.

Data centres are highly sensitive to voltage and current fluctuations, and power consumption and generation issues, driving the demand for electronic test equipment.

802.11ax Wi-Fi standard. Continuous evolution of communications standards boosts the demand for test equipment as more standards increase the number of test cases and add to the difficulty in performing measurements. For example, 802.11ax is providing growth for T&M vendors due to specific features requiring high RF performance and greater measurement accuracy.

The IoT device and network test equipment

Interconnectivity of consumer electronics has led to complex test scenarios that require new and scalable test equipment. Narrowband IoT (NB-IoT) and mobile IoT standards such as Bluetooth Low Energy (BLE) and LTE-M are creating a new part of the market that did not exist earlier.

The IoT is generating a growing need for inspecting, testing and verifying smart products and other equipment to ensure that these work correctly once deployed. This is driving the need for more field data acquisition (DAQ) systems.

This data can also be used to fine-tune designs in laboratories. With the IoT, DAQ has the potential to be put on a large number of devices to generate data on how it is used. This data would be sent to a server and data analytics application, and could be used by an engineer working on the design of the next generation of that product.

The IoT is taking us into a new world of connected devices, which is enabled by the network infrastructure, the Internet and World Wide Web. To prevent network security breaches, the network stack, security layers and firewalls on these devices need to be tested by simulating various use-cases.

The IoT devices are tested across different cloud IoT platforms to ensure their effective usability. All connected devices need to be tested across various protocols when there are multiple components in an IoT system. Different components in such a scenario will use different protocols to communicate with each other. In IoT testing, it is critical to test passwords and credentials, data interfaces